RESEARCH REPORT

2D PROJECTILES SIMULATION

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Abstract
nothing yet

Contents

0.1 Introduction

The aim of this report is to document the process of my interpretation and derivation of the fundamnetal laws of motions. More specifically, the projectile motions of a simple object. I intend to document each step of my simulation and formula derivation to deepen my understanding of the mechanics of motions in the physical reality.

0.2 Fundamental Formulas Of Motions

The following formulas is used to predict an object's acceleration, time, displacement and etc, by providing a combination of values from other variables.

$$v = u + at \tag{1}$$

$$s = ut + \frac{1}{2}at^2 \tag{2}$$

$$v^2 = u^2 + 2as \tag{3}$$

0.2.1 Derivation of the first formula

Acceleration is defined as the rate of change in velocity. In other words, how fast the speed of the object changes. To measure this, We take the difference between the final velocity with the initial velocity, divided by time, to see how much per second, does the speed change over time. As such, the formula is as follows:

$$a = \frac{v - u}{t} \tag{4}$$

The derivation of the first formula is simply a rearrangement of the acceleration formula with velocity as the subject.

$$a = \frac{v - u}{t} \tag{5}$$

$$at = v - u \tag{6}$$

$$v = u + at \tag{7}$$

0.2.2 Derivation of the second formula

After obtaining the first formula, we can substitute the formula (1) into formula (2).

$$v = u + at \tag{8}$$

$$s = \frac{v + u}{2}t \tag{9}$$

$$\tag{10}$$

The algebraic simplification is as follows.

$$
\begin{aligned}
\mathrm{s} &= \frac{u+v}{2}(t) \\
\mathrm{s} &= \frac{u+(u+at)}{2}(t) \\
\mathrm{s} &= \frac{2u+at}{2}(t) \\
\mathrm{s} &= (\frac{2u}{2}+\frac{at}{2})(t) \\
\mathrm{s} &= (ut+\frac{1}{2})(at^2)
\end{aligned} \tag{11}
$$

0.2.3 Derivation of the third formula

The third formula has

$$
\begin{aligned}
\mathrm{t} &= \frac{v-u}{a} \\
\mathrm{s} &= \frac{v+u}{2}\left(\frac{v-u}{a}\right) \\
\mathrm{s} &= \frac{v^2-vu+vu-u^2}{2a} \\
\mathrm{s} &= \frac{v^2-u^2}{2a} \\
2\mathrm{as} &= v^2+u^2 \\
\mathrm{v}^2 &= u^2+2as
\end{aligned} \tag{12}
$$